

Technical Document #158: Cloud Computing - Comprehensive Overview

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Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Multi-cloud strategies use services from multiple cloud providers. They reduce vendor lock-in and optimize for cost and performance.

Content delivery networks (CDNs) distribute content globally to reduce latency. CloudFront and Cloudflare are popular CDN providers.

Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files. Terraform and CloudFormation are popular IaC tools.

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. AWS EC2 and Azure VMs are popular IaaS offerings.

Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Key Takeaways:

- Kubernetes orchestrates containerized applications across clusters of machines.
- Content delivery networks (CDNs) distribute content globally to reduce latency.
- Platform as a Service (PaaS) provides a platform for developing and deploying applications.